NCEA Level 2 Mathematics

91262 · Apply calculus methods in solving problems (Calculus)

Excellence-level practice questions

PRACTICE QUESTIONS

About this paper

- This is a **practice past paper**, written in the style of NCEA excellence questions. It is **not** an official NZQA paper, and it may or may not be similar to the actual exam.
- Questions aim at the excellence end of the standard, so expect them to be harder than an average exam question. Don't be discouraged if some take a while.
- **Questions 21 onwards are a bit harder** than the earlier questions in this paper.

How to use it

- Try each question on your own first, ideally under timed, exam-like conditions.
- Show full working and give clear final answers, as you would in the real exam.
- Only check the answers once you've made a proper attempt.

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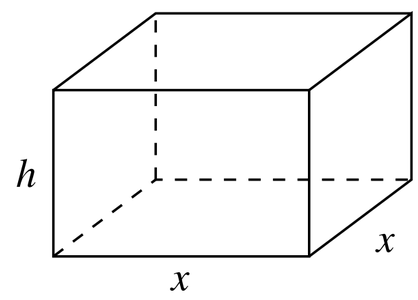
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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

1. An open box (i.e. a box with a base and four sides, but no lid) has a square base with sides of length x cm, and a height of h cm.



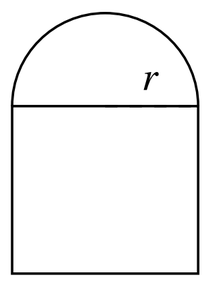
The box is made from 300 cm^2 of thin sheet metal.

Find the maximum possible volume of the box.

You must use calculus and show any derivatives that you need to find when solving this problem.

Justify that your answer is a maximum.

2. A window is in the shape of a rectangle with a semi-circle on top, as shown below. The radius of the semi-circle is r metres.



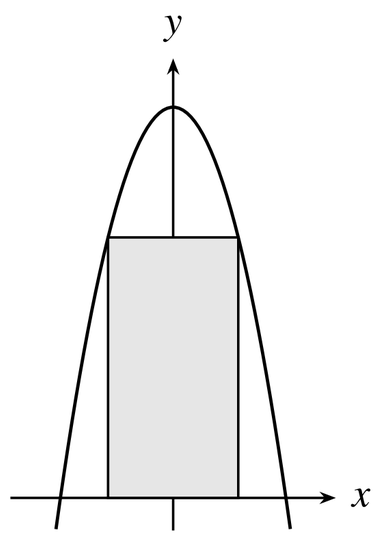
The total perimeter of the window frame (including the base, but not the line between the rectangle and the semi-circle) is 6 m.

Find the radius that lets the greatest amount of light through the window (i.e. that maximises the area of the window), and find this maximum area.

You must use calculus and show any derivatives that you need to find when solving this problem.

(Circumference of a circle = $2\pi r$, Area of a circle = πr^2)

3. A rectangle is drawn with two vertices on the x-axis, and the other two vertices on the parabola $y = 12-x^2$, as shown below.



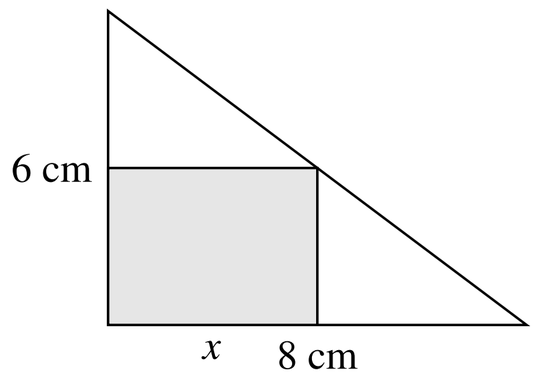
Find the maximum possible area of the rectangle.

You must use calculus and show any derivatives that you need to find when solving this problem.

Justify that your answer is a maximum.

4. A rectangle is drawn inside a right-angled triangle, so that two sides of the rectangle lie along the two shorter sides of the triangle, and one vertex of the rectangle lies on the hypotenuse.

The two shorter sides of the triangle have lengths 8 cm and 6 cm.



Find the maximum possible area of the rectangle.

You must use calculus and show any derivatives that you need to find when solving this problem.

5. A farmer has 240 m of fencing to build a rectangular enclosure. The enclosure is divided into three equal pens by two internal fences that are parallel to the width, as shown below.

Diagram is NOT to scale

All 240 m of fencing is used.

Find the dimensions of the enclosure that give the maximum total area, and state this area.

You must use calculus and show any derivatives that you need to find when solving this problem.

6. If $x + 2y = 10$, use calculus to find the minimum value of $x^2 + y^2$.

Justify that your answer is a minimum.

7. A closed cylindrical tin must have a volume of 1000 cm^3.

The material for the top and the bottom of the tin costs twice as much per square centimetre as the material for the curved side.

Find the radius of the tin that will minimise the cost of the material.

You must use calculus and show any derivatives that you need to find when solving this problem.

(Volume of a cylinder = $\pi r^2 h$, Curved surface area of a cylinder = $2\pi r h$)

8. A particle moves along a straight line. Its acceleration, a m s^{-2}, t seconds after it passes a fixed point O, is given by

$$a(t) = 6t - 12$$

When it passes O, the particle has a velocity of 9 m s^{-1}.

Find the displacement of the particle from O, 4 seconds after it passes O.

You must use calculus and show any derivatives that you need to find when solving this problem.

9. A truck passes a set of traffic lights, travelling at a constant speed of $12\ \mathrm{m\,s^{-1}}$.

5 seconds later, a motorbike starts from rest at the traffic lights, and accelerates at a constant $2\ \mathrm{m\,s^{-2}}$ in the same direction as the truck.

How far from the traffic lights is the motorbike when it catches up to the truck, and what is its speed at that moment?

You must use calculus and show any derivatives that you need to find when solving this problem.

10. A ball is thrown vertically upwards from the edge of the roof of a building that is 25 m high. The ball leaves the thrower's hand with a velocity of 20 $\mathrm{m\,s^{-1}}$.

The ball's acceleration is a constant $-9.8\ \mathrm{m\,s^{-2}}$.

The ball misses the roof on its way down, and falls to the ground beside the building.

Find the speed of the ball when it hits the ground.

You must use calculus and show any derivatives that you need to find when solving this problem.

11. A driver is travelling at 25 $\mathrm{m\,s^{-1}}$ when she sees a hazard on the road 80 m ahead.

It takes her 0.8 seconds to react before she applies the brakes. The brakes then decelerate the car at a constant rate of 5 $\mathrm{m\,s^{-2}}$.

(i) Will the car stop before it reaches the hazard? Justify your answer.

(ii) Find the minimum constant deceleration that the brakes would need to provide for the car to stop exactly at the hazard.

You must use calculus and show any derivatives that you need to find when solving this problem.

12. The line $y = 4x + c$ is a tangent to the curve $y = x^3 - 2x^2 + 1$.

Find the possible values of c.

You must use calculus and show any derivatives that you need to find when solving this problem.

13. Two tangents are drawn from the point $(1, -2)$ to the curve $y = x^2 - 4x + 5$.

Find the equation of each tangent.

You must use calculus and show any derivatives that you need to find when solving this problem.

14. The tangent to the curve $y = x^3 - 3x^2 + 2$ at the point where $x = -1$ meets the curve again at the point P.

Find the coordinates of P.

You must use calculus and show any derivatives that you need to find when solving this problem.

15. A cubic function $f(x) = ax^3 + bx^2 + cx + d$:

- passes through the origin
- has a gradient of 3 at the origin
- has a stationary point at $(2, -4)$.

Find the values of a, b, c, and d, and determine the nature of the stationary point at $(2, -4)$.

Justify your answer.

16. The curve $y = x^3 + px^2 + qx + 4$ has a stationary point at $(1, 2)$.

Find the values of p and q, and find the coordinates and nature of the other stationary point.

You must use calculus and show any derivatives that you need to find when solving this problem.

17. The gradient function of a curve is given by

$$\frac{dy}{dx} = 3x^2 + kx - 4$$

where k is a constant. The curve has a turning point when $x = 2$, and passes through the point $(0, 5)$.

Find the equation of the curve, and the coordinates of its other turning point.

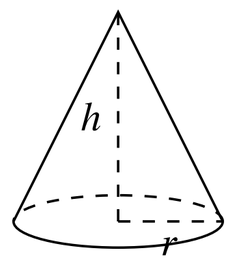
18. Find the range of values of k for which the function

$$f(x) = x^3 + kx^2 + 12x - 5$$

is increasing for all values of x.

You must use calculus and show any derivatives that you need to find when solving this problem.

19. Sand is poured onto the ground, forming a cone. The height of the cone is always twice the radius of its base.



Find the rate of change of the volume of the cone with respect to its radius, at the instant when the volume of the cone is 18π cm^3.

(Volume of a cone = $\frac{1}{3}\pi r^2 h$)

20. A curve has the equation $y = x^3 - 3k^2x + 4$, where k is a positive constant.

(i)Find, in terms of k, the range of values of x for which the curve is decreasing.

(ii)Find the value of k for which the local minimum of the curve lies on the x-axis.

You must use calculus and show any derivatives that you need to find when solving this problem.

21. Use first principles to find the derivative of

$$f(x) = x^3 - 2x$$

(First principles: $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$)

22. The normal to a curve at a point is the line through that point that is perpendicular to the tangent.

Find the equation of the normal to the curve $y = x^2 - 4x$ at the point $(1, -3)$, and find the coordinates of the other point where this normal meets the curve.

You must use calculus and show any derivatives that you need to find when solving this problem.

23. The tangent to the curve $y = 8 - x^2$ at the point where $x = 2$ forms a triangle with the x-axis and the y-axis.

Find the area of this triangle.

You must use calculus and show any derivatives that you need to find when solving this problem.

24. For the curve $y = x^3 - 6x^2 + 9x + 1$, the gradient of the curve changes as x changes.

Find the smallest possible value of the gradient of the curve, and the point on the curve where this occurs.

You must use calculus and show any derivatives that you need to find when solving this problem.

25. Two particles, A and B, move along the same straight line. Their displacements from a fixed point O, in metres, t seconds after they start moving are given by

$$s_A = t^2 - 4t \qquad \text{and} \qquad s_B = 6t - t^2$$

(i) At what time are the two particles moving with the same velocity? How far apart are they at this time?

(ii) When, after they start, are the two particles at the same position?

You must use calculus and show any derivatives that you need to find when solving this problem.

26. P is the point on the curve $y = x^2$ that is closest to the point $\left(16, \frac{1}{2}\right)$.

By minimising the square of the distance between the two points, find the coordinates of P, and the shortest distance.

You must use calculus and show any derivatives that you need to find when solving this problem.

(Distance between two points: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$)

27. A small business makes and sells x items each day.

The cost of making x items, in dollars, is given by

$$C(x) = x^3 - 24x^2 + 200x$$

Each item is sold for \$155.

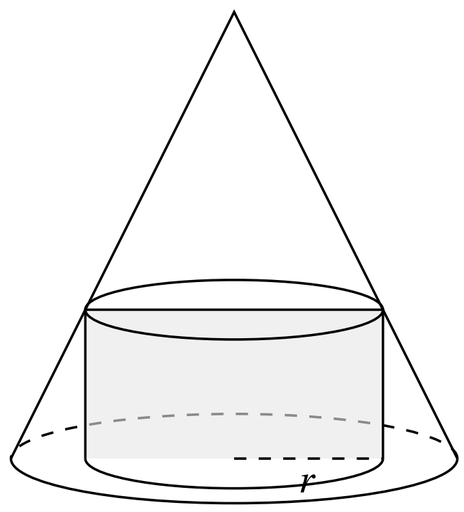
Find the number of items the business should make each day to maximise its profit, and find this maximum profit.

You must use calculus and show any derivatives that you need to find when solving this problem.

Justify that your answer is a maximum.

28. A cylinder is placed inside a cone so that the base of the cylinder sits on the base of the cone, and the top edge of the cylinder touches the cone.

The cone has a base radius of 6 cm and a height of 12 cm.



Find the radius of the cylinder that gives it the maximum possible volume, and find this volume.

You must use calculus and show any derivatives that you need to find when solving this problem.

(Volume of a cylinder = $\pi r^2 h$)

29. The gradient of the gradient function of a curve $y = f(x)$ is given by $6x - 4$.

The curve has a turning point at $(1, 2)$.

Find the equation of the curve.

30. A function $y = f(x)$ has all of the following properties:

- $f'(x) > 0$ for $x < -1$ and for $x > 3$
- $f'(x) < 0$ for $-1 < x < 3$
- $f'(-1) = f'(3) = 0$
- $f(-1) = 4$, $f(0) = 2$, and $f(3) = -3$.

Sketch a possible graph of $y = f(x)$, and of its gradient function $y = f'(x)$.

Label the turning points of $f(x)$, and explain how each feature of your gradient graph matches the properties above.

31. Show that no tangent to the curve $y = x^3 + 3x$ has a gradient less than 3.

Hence find the equation of the tangent to the curve that has the smallest possible gradient.

32. The points $A(a, a^2)$ and $B(b, b^2)$ lie on the curve $y = x^2$.

The tangents to the curve at A and at B are perpendicular to each other.

Show that these two tangents always meet on the line $y = -\frac{1}{4}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

33. The perimeter of a square is P cm, and its area is A cm^2.

Find the rate of change of the area of the square with respect to its perimeter, when the perimeter is 20 cm.

34. A gutter is made from a long rectangular sheet of metal that is 30 cm wide. The two edges of the sheet are folded up at right angles to form the sides of the gutter, as shown in the cross-section below.



Diagram is NOT to scale

Find the height, x cm, of the sides that will let the gutter carry the most water (i.e. that maximises the area of the cross-section).

You must use calculus and show any derivatives that you need to find when solving this problem.

35. Find the range of values of k for which the equation

$$x^3 - 3x^2 = k$$

has three distinct real solutions.

You must use calculus, and explain your reasoning with reference to the graph of $y = x^3 - 3x^2$.

36. A chord is drawn between the points on the curve $y = x^3 - x$ where $x = 0$ and $x = 3$.

Find the coordinates of the point(s) on the curve, between $x = 0$ and $x = 3$, where the tangent is parallel to this chord.

You must use calculus and show any derivatives that you need to find when solving this problem.

37. A lift starts from rest, and accelerates upwards at $1.2\ \text{m}\,\text{s}^{-2}$ for 5 seconds.

It then travels at a constant speed, before decelerating at $1.5\ \text{m}\,\text{s}^{-2}$ until it comes to rest.

The lift travels a total distance of 60 m.

For how many seconds does the lift travel at a constant speed?

You must use calculus and show any derivatives that you need to find when solving this problem.

38. A quadratic function $y = f(x)$ has a maximum turning point at $(3, 2)$.

The tangent to the graph of $f(x)$ at the point where $x = 1$ passes through the origin.

Find the equation of $f(x)$.

You must use calculus and show any derivatives that you need to find when solving this problem.

39. A boat makes a 300 km trip at a constant speed of v km h^{-1}.

The total cost of the trip, in dollars, is given by

$$C = 300\left(0.02v^2 + \frac{40}{v}\right)$$

Find the speed that minimises the total cost of the trip, and find this minimum cost.

You must use calculus and show any derivatives that you need to find when solving this problem.

Justify that your answer is a minimum.

40. A particle starts at a fixed point O and moves along a straight line. Its velocity, v m s^{-1}, t seconds after it leaves O, is given by

$$v = 12 - 3t^2$$

(i)Find the greatest distance of the particle from O in the positive direction.

(ii)Find the time when the particle returns to O.

You must use calculus and show any derivatives that you need to find when solving this problem.